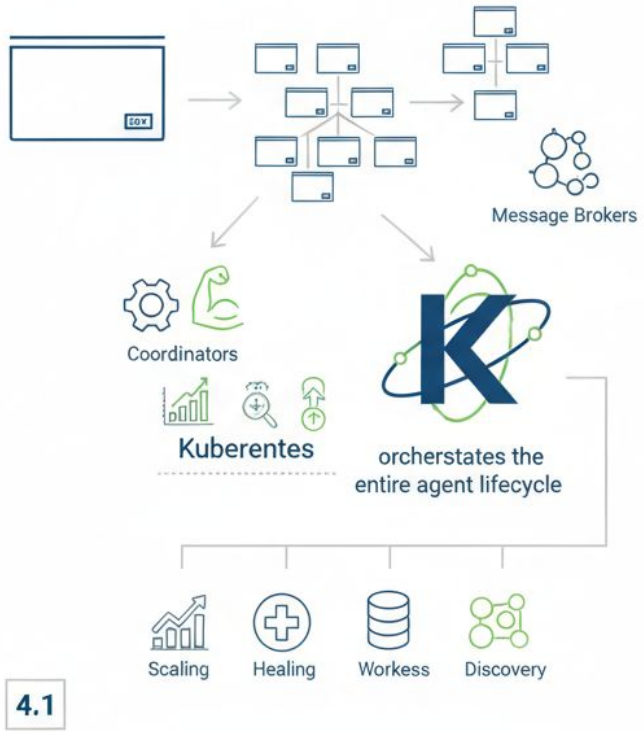
U.S. General Services Administration

Chapter 4.3: Container Orchestration

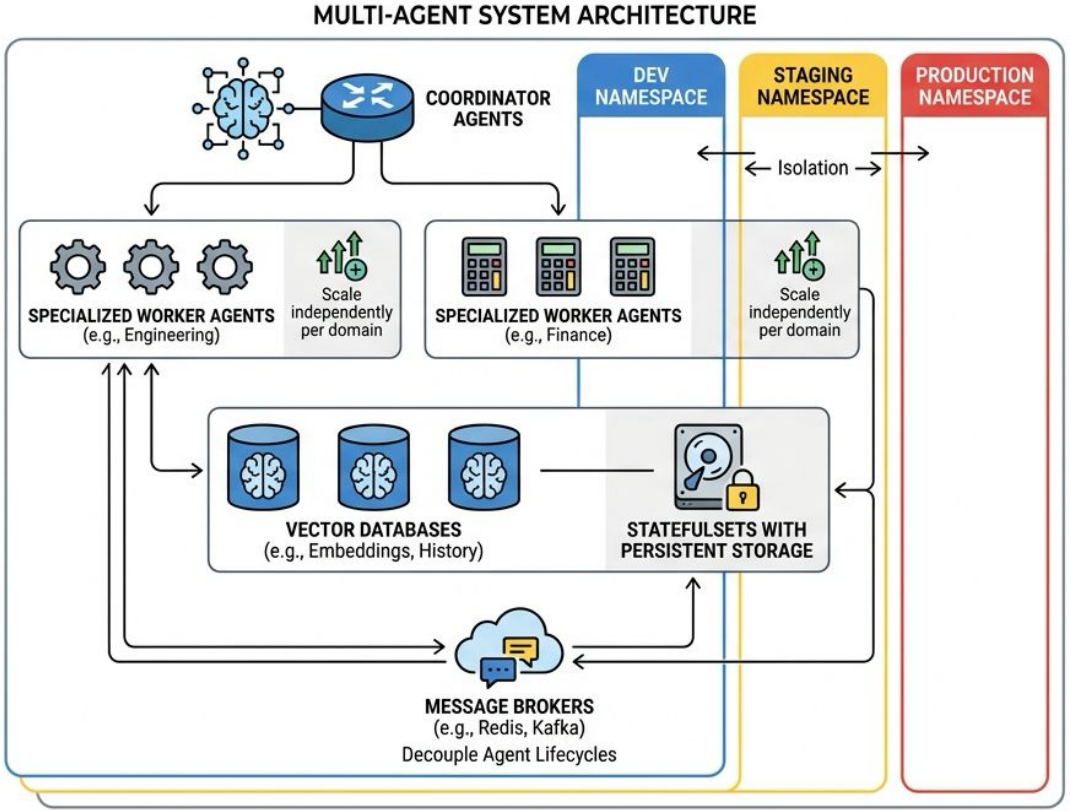
US General Service Administration

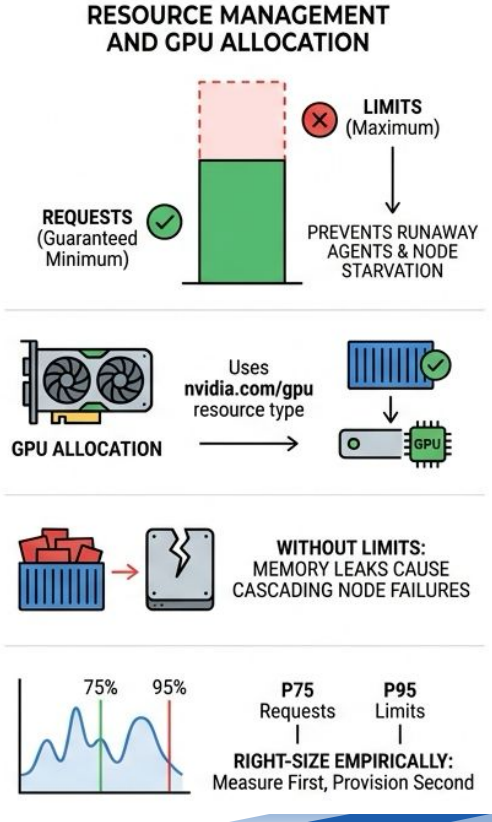
AI Community of Practice - March 2026

Declarative Deployments and Desired State

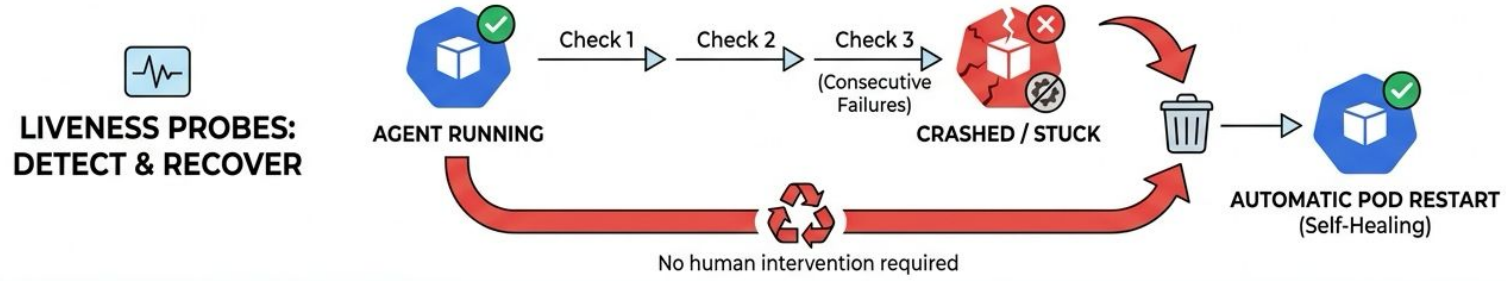


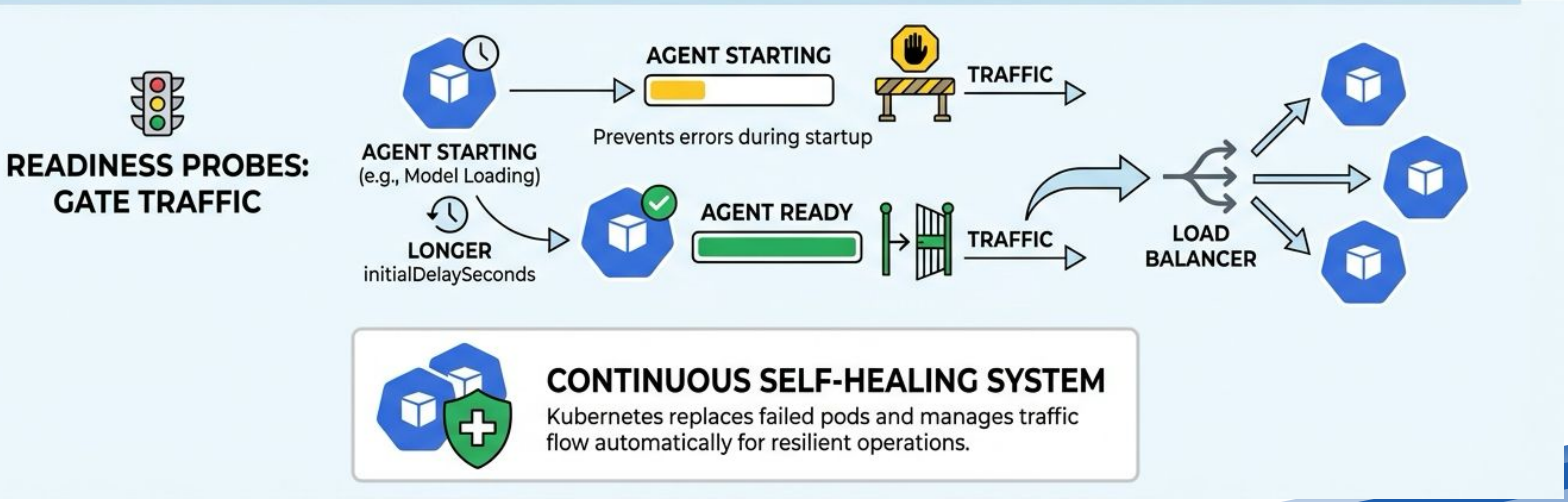
Multi-Agent System Architecture in Kubernetes



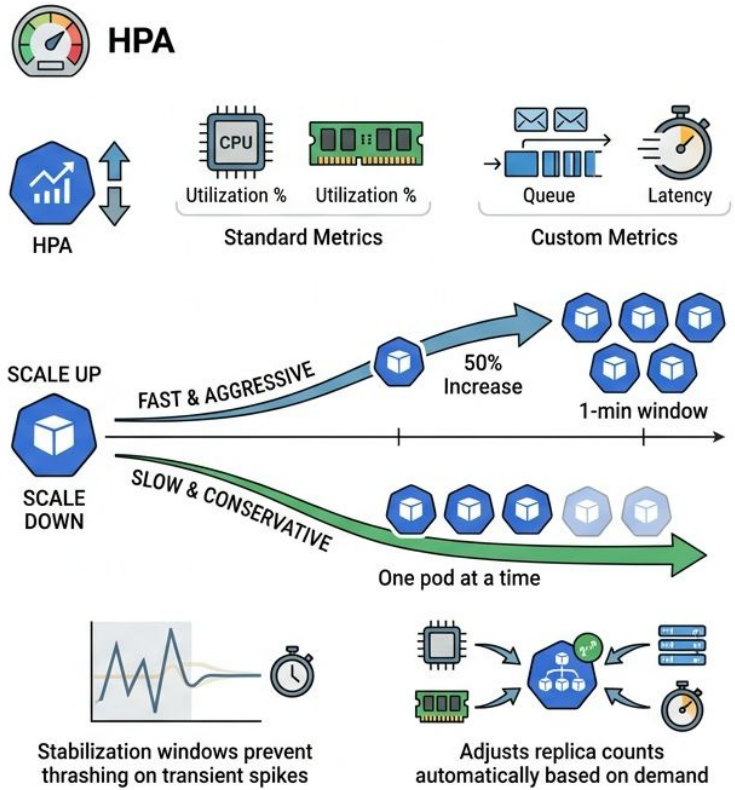


Health Probes and Self-Healing



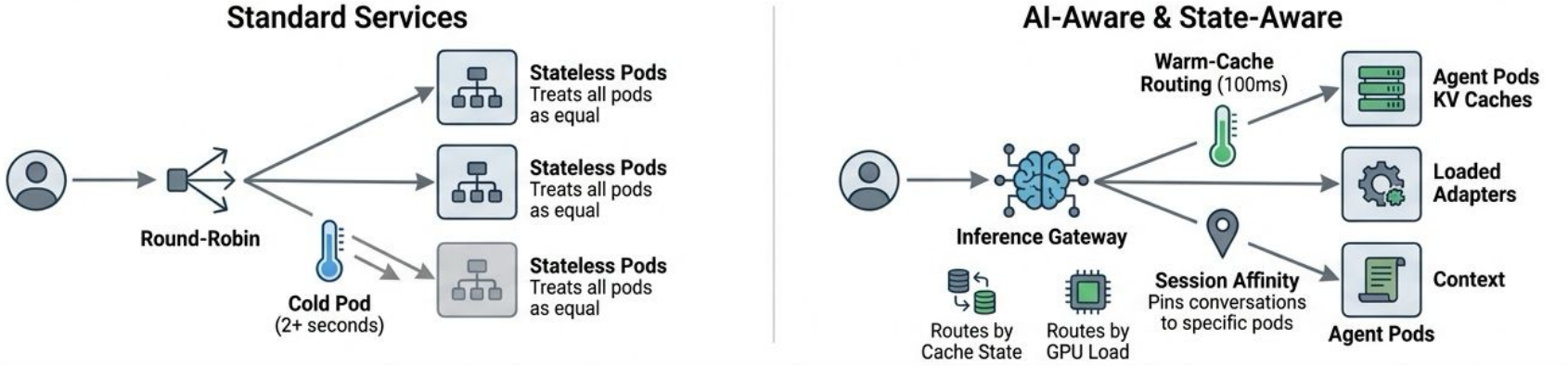


Horizontal Pod Autoscaler and Scaling Policies



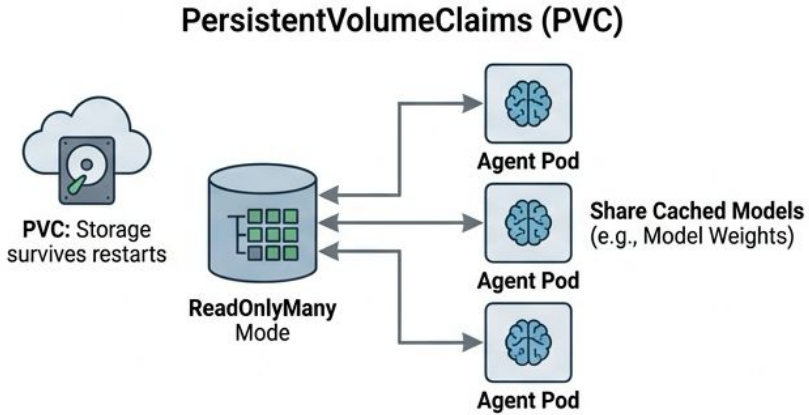
- HPA monitors metrics and adjusts replica counts automatically
- Standard metrics: CPU utilization and memory usage
- Custom metrics: queue depth, request latency
- Scale up fast and aggressively; scale down slowly
- Stabilization windows prevent thrashing on transient spikes

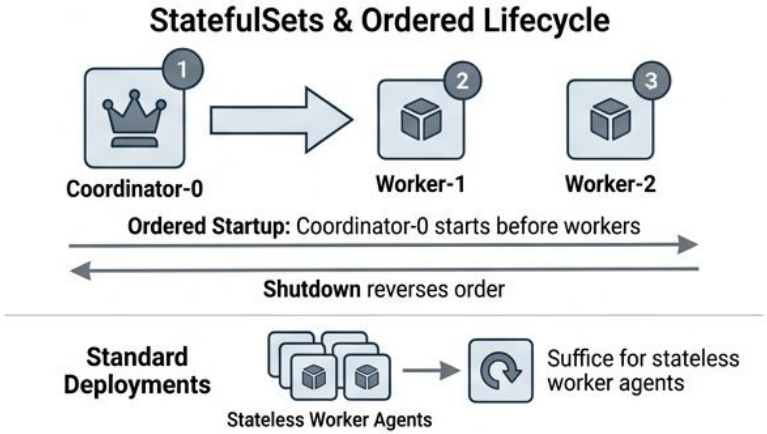
AI-Aware Load Balancing



- Standard Services use round-robin, treating all pods as equal
- Agent pods carry state: KV caches, loaded adapters, context
- Warm-cache routing delivers 100ms; cold pod takes 2+ seconds
- Inference gateways route by cache state and GPU load
- Session affinity pins conversations to specific pods

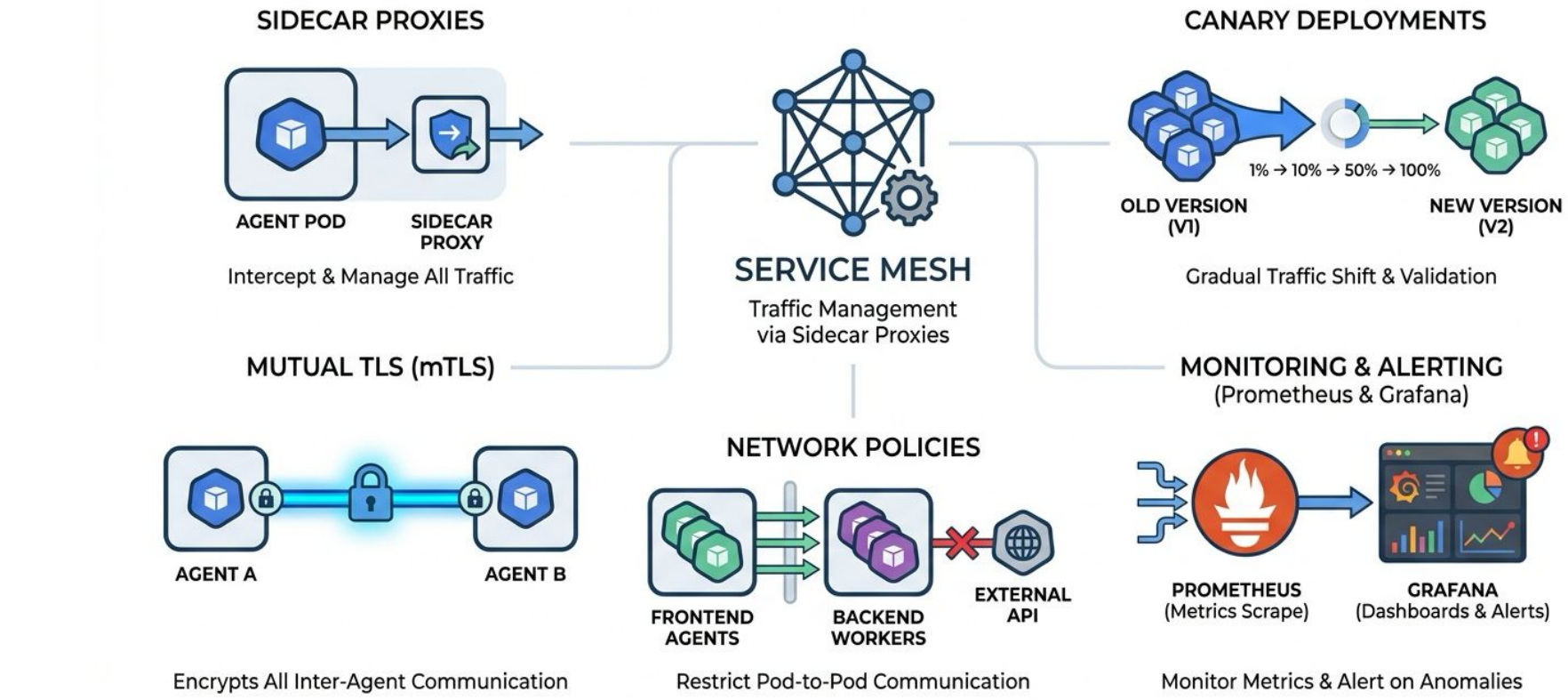
Persistent Storage and StatefulSets



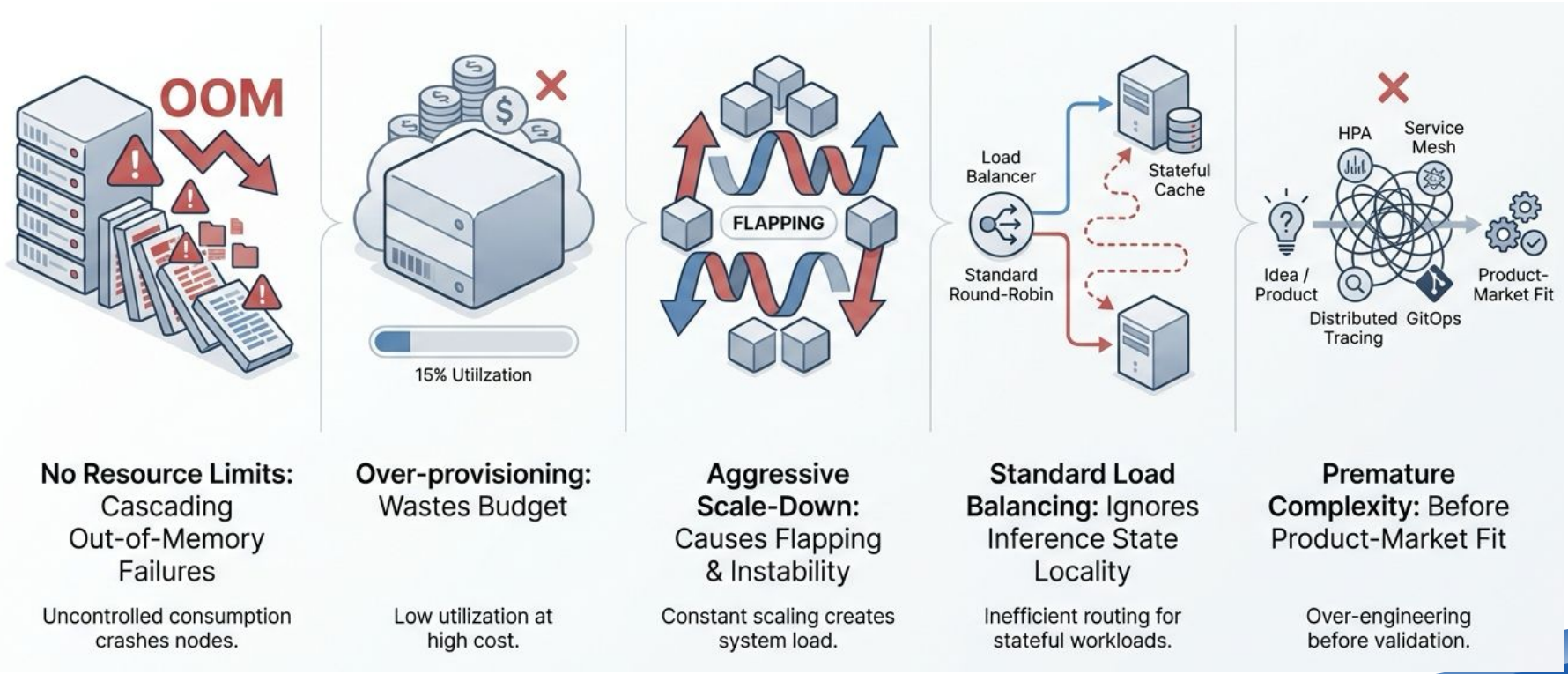


- PersistentVolumeClaims survive pod restarts and reschedules
- ReadOnlyMany mode lets all pods share cached models
- StatefulSets provide stable identity and ordered startup
- Coordinator-0 starts before workers; shutdown reverses order
- Standard Deployments suffice for stateless worker agents

Service Mesh and Production Observability



Common Pitfalls in Production



No Resource Limits: Cascading Out-of-Memory Failures

Uncontrolled consumption crashes nodes.

Over-provisioning: Wastes Budget

Low utilization at high cost.

Aggressive Scale-Down: Causes Flapping & Instability

Constant scaling creates system load.

Standard Load Balancing: Ignores Inference State Locality

Inefficient routing for stateful workloads.

Premature Complexity: Before Product-Market Fit

Over-engineering before validation.

Worked Example -- Customer Service Agent System

- Coordinator: 2 replicas, 0.5 CPU, 1GB RAM, routing logic
- Technical worker: 3 replicas, GPU-equipped, 8GB RAM
- HPA scales workers from 3 to 20 based on queue depth
- PVC shares 100GB model cache across all worker pods
- LoadBalancer Service exposes coordinator to external traffic

Key Takeaways

- Kubernetes transforms manual ops into declarative desired state
- Resource requests, limits, and probes are non-negotiable
- Asymmetric scaling: fast up, slow down prevents flapping
- Inference workloads need cache-aware routing, not round-robin
- Next: Chapter 4.4 tackles performance profiling and optimization